Marcelo Araújo Silva

Full-Stack Developer | Node.js / NestJS / TypeScript

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Summary

Full-Stack Developer focused on backend, with a background in Computational Biology. I build scalable applications with **Node.js/NestJS** and **TypeScript** on the backend and **React/Next.js** on the frontend, with hands-on experience in **REST APIs**, **PostgreSQL**, and **Prisma ORM**. Remote-ready from GMT-3 with strong overlap with European mornings, I pair scientific research rigor — including an international peer-reviewed publication — with daily software delivery.

Technologies

Languages	JavaScript/TypeScript, Python, Java, R, Bash
Backend	Node.js, NestJS, Express, Spring Boot, REST APIs, Prisma ORM
Frontend	React, Next.js
Data & Infra	PostgreSQL, Docker, Git/GitHub, Linux, Conda, Nextflow

Experience

Backend Developer — TITAN Apr 2026 – Present

- Promoted from Trainee to Backend Developer in July 2026 (currently part-time), following a technical training program.
- Develop backend applications and **REST APIs** in **Node.js/NestJS** with **Prisma ORM**, including user authentication flows.

Full-Stack Developer — Aletheia (Freelance) 2026

- Developed the registration and account-administration system for the **Bahia Biology Olympiad**, serving **hundreds of teachers and students** and automating exam scheduling/delivery and registrant management.
- Implemented authentication and account-security flows in **Node.js/Express** with **Prisma ORM** and **PostgreSQL**.
- Built the **REST API** for user credentialing, including PDF report generation and transactional emails (Handlebars) delivered via SMTP.
- Developed the frontend in **React/TypeScript** and led the technical planning of the solution.

Data Scientist — Bioinformatics Lab (UFBA) Research Internship - 2022 – 2025

- Processed and performed quality control on **2,650+ samples** on Linux servers (**6+ TB**), generating standardized reports and outputs for downstream analysis.
- Developed modular tools in **Python** (scripts and CLIs) to automate multi-step workflows with clear input/output handling, focusing on maintainability and reproducibility.
- Conducted statistical analyses and visualizations using **R/Python** for scientific validation and communication (publication-ready figures and analytical narratives), supporting study conclusions and decision-making.
- Maintained version-controlled repositories (**Git/GitHub**) and technical documentation for collaboration and result traceability.

Software Engineering Projects

BioPoints — 1st Place, LIAO Hackathon @ UFBA (Full-Stack + ML) 2026

- Won **1st place** in a university hackathon themed around climate change, smart cities, and applied AI; conceived the product and led delivery end-to-end under strict time constraints.
- Architected and shipped the full **backend** (Node.js, NestJS, Prisma ORM, PostgreSQL): REST API, relational schema modeling multi-actor B2B workflows, and an ML-model serving endpoint.
- Built the **frontend** (React, Next.js) with an interactive interface that consumes the backend API in real time for on-the-fly computation and visualization.
- Authored the pitch deck and delivered the live presentation that secured the win against **[N] competing** university teams.

Weather Data (Spring Boot, PostgreSQL, React) 2026 – Present

- Built a full-stack system that analyzes weather conditions on a specific date (e.g., the user's birth date) by integrating external geolocation and historical climate data APIs.
- Implemented a **REST API** with PostgreSQL persistence, consuming Open-Meteo and geocoding APIs while following repository/service layers, DTOs, and centralized exception handling.
- Developed climate analysis logic on the backend (averages, maximums, minimums, and textual classifications) and data visualization with dynamic charts on the React frontend using Recharts.

Idle Game Desktop (React + Electron) 2025 – Present

- Developed a desktop application with React (renderer) and Electron (main), focusing on state organization, UI, and integration with the Node.js runtime.
- Implemented data collection and JSON normalization in the main process and used IPC for stable inter-process communication.

Education

2026 – Present **Postgraduate Program in Cybersecurity** and Data Governance — PUC Minas
2026 – Present **B.S. in Information Systems** — State University of Bahia (UNEB), Salvador-BA, Brazil
2021 – 2025 **B.S. in Biological Sciences** — Federal University of Bahia (UFBA), Salvador-BA, Brazil

Publications

2026 **Mycological Progress** — *Taxonomic and phylogenetic studies reveal three new species of the genus Geastrum in the Corollina section from Brazil.* Second author. DOI: 10.1007/s11557-025-02118-8

Certifications

2024 Duolingo English Test — Overall Score: 135
2023 Introduction to Computing — Institute of Exact Sciences, UFMG

Languages

Portuguese Native
English Advanced
Spanish Basic

Open Source Contribution

2026 Open source contribution — Curation and analysis of a public dataset in Python (Brazilian companies and share capital), with version control and documentation on GitHub.